

CATEGORICAL REPARAMETERIZATION WITH GUMBEL-SOFTMAX

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ABSTRACT

Categorical variables are a natural choice for representing discrete structure in the world. However, stochastic neural networks rarely use categorical latent variables due to the inability to backpropagate through samples. In this work, we present an efficient gradient estimator that replaces the non-differentiable sample from a categorical distribution with a differentiable sample from a novel Gumbel-Softmax distribution. This distribution has the essential property that it can be smoothly annealed into a categorical distribution. We show that our Gumbel-Softmax estimator outperforms state-of-the-art gradient estimators on structured output prediction and unsupervised generative modeling tasks with categorical latent variables, and enables large speedups on semi-supervised classification.

1 INTRODUCTION

Stochastic neural networks with discrete random variables are a powerful technique for representing distributions encountered in unsupervised learning, language modeling, attention mechanisms, and reinforcement learning domains. For example, discrete variables have been used to learn probabilistic latent representations that correspond to distinct semantic classes (Kingma et al., 2014), image regions (Xu et al., 2015), and memory locations (Graves et al., 2014; Graves et al., 2016). Discrete representations are often more interpretable (Chen et al., 2016) and more computationally efficient (Rae et al., 2016) than their continuous analogues.

However, stochastic networks with discrete variables are difficult to train because the backpropagation algorithm — while permitting efficient computation of parameter gradients — cannot be applied to non-differentiable layers. Prior work on stochastic gradient estimation has traditionally focused on either score function estimators augmented with Monte Carlo variance reduction techniques (Paisley et al., 2012; Mnih & Gregor, 2014; Gu et al., 2016; Gregor et al., 2013), or biased path derivative estimators for Bernoulli variables (Bengio et al., 2013). However, no existing gradient estimator has been formulated specifically for categorical variables. The contributions of this work are threefold:

1. We introduce Gumbel-Softmax, a continuous distribution on the simplex that can approximate categorical samples, and whose parameter gradients can be easily computed via the reparameterization trick.
2. We show experimentally that Gumbel-Softmax outperforms all single-sample gradient estimators on both Bernoulli variables and categorical variables.
3. We show that this estimator can be used to efficiently train semi-supervised models (e.g. Kingma et al. (2014)) without costly marginalization over unobserved categorical latent variables.

The practical outcome of this paper is a simple, differentiable approximate sampling mechanism for categorical variables that can be integrated into neural networks and trained using standard backpropagation.

*Work done during an internship at Google Brain.

2 THE GUMBEL-SOFTMAX DISTRIBUTION

We begin by defining the Gumbel-Softmax distribution, a continuous distribution over the simplex that can approximate samples from a categorical distribution. Let z be a categorical variable with class probabilities $\pi_1, \pi_2, \dots, \pi_k$. For the remainder of this paper we assume categorical samples are encoded as k -dimensional one-hot vectors lying on the corners of the $(k-1)$ -dimensional simplex, Δ^{k-1} . This allows us to define quantities such as the element-wise mean $\mathbb{E}_p[z] = [\pi_1, \dots, \pi_k]$ of these vectors.

The Gumbel-Max trick (Gumbel, 1954; Maddison et al., 2014) provides a simple and efficient way to draw samples z from a categorical distribution with class probabilities π :

$$z = \text{one_hot} \left(\arg \max_i [g_i + \log \pi_i] \right) \quad (1)$$

where $g_1 \dots g_k$ are i.i.d samples drawn from $\text{Gumbel}(0, 1)$ ¹. We use the softmax function as a continuous, differentiable approximation to $\arg \max$, and generate k -dimensional sample vectors $y \in \Delta^{k-1}$ where

$$y_i = \frac{\exp((\log(\pi_i) + g_i)/\tau)}{\sum_{j=1}^k \exp((\log(\pi_j) + g_j)/\tau)} \quad \text{for } i = 1, \dots, k. \quad (2)$$

The density of the Gumbel-Softmax distribution (derived in Appendix B) is:

$$p_{\pi, \tau}(y_1, \dots, y_k) = \Gamma(k) \tau^{k-1} \left(\sum_{i=1}^k \pi_i / y_i^\tau \right)^{-k} \prod_{i=1}^k (\pi_i / y_i^{\tau+1}) \quad (3)$$

This distribution was independently discovered by Maddison et al. (2016), where it is referred to as the concrete distribution. As the softmax temperature τ approaches 0, samples from the Gumbel-Softmax distribution become one-hot and the Gumbel-Softmax distribution becomes identical to the categorical distribution $p(z)$.

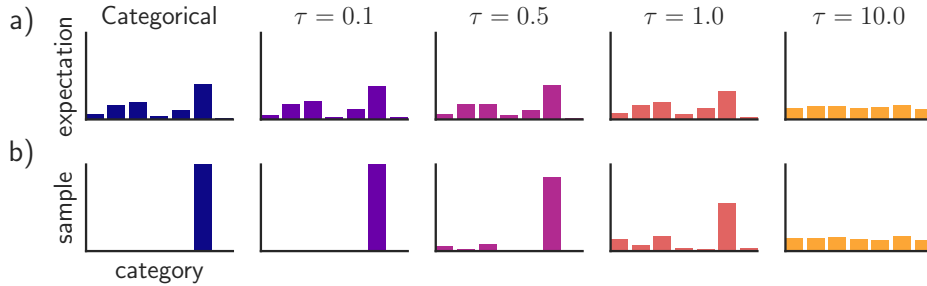


Figure 1: The Gumbel-Softmax distribution interpolates between discrete one-hot-encoded categorical distributions and continuous categorical densities. (a) For low temperatures ($\tau = 0.1, \tau = 0.5$), the expected value of a Gumbel-Softmax random variable approaches the expected value of a categorical random variable with the same logits. As the temperature increases ($\tau = 1.0, \tau = 10.0$), the expected value converges to a uniform distribution over the categories. (b) Samples from Gumbel-Softmax distributions are identical to samples from a categorical distribution as $\tau \rightarrow 0$. At higher temperatures, Gumbel-Softmax samples are no longer one-hot, and become uniform as $\tau \rightarrow \infty$.

2.1 GUMBEL-SOFTMAX ESTIMATOR

The Gumbel-Softmax distribution is smooth for $\tau > 0$, and therefore has a well-defined gradient $\partial y / \partial \pi$ with respect to the parameters π . Thus, by replacing categorical samples with Gumbel-Softmax samples we can use backpropagation to compute gradients (see Section 3.1). We denote

¹The $\text{Gumbel}(0, 1)$ distribution can be sampled using inverse transform sampling by drawing $u \sim \text{Uniform}(0, 1)$ and computing $g = -\log(-\log(u))$.

this procedure of replacing non-differentiable categorical samples with a differentiable approximation during training as the Gumbel-Softmax estimator.

While Gumbel-Softmax samples are differentiable, they are not identical to samples from the corresponding categorical distribution for non-zero temperature. For learning, there is a tradeoff between small temperatures, where samples are close to one-hot but the variance of the gradients is large, and large temperatures, where samples are smooth but the variance of the gradients is small (Figure 1). In practice, we start at a high temperature and anneal to a small but non-zero temperature.

In our experiments, we find that the softmax temperature τ can be annealed according to a variety of schedules and still perform well. If τ is a learned parameter (rather than annealed via a fixed schedule), this scheme can be interpreted as entropy regularization (Szegedy et al., 2015; Pereyra et al., 2016), where the Gumbel-Softmax distribution can adaptively adjust the “confidence” of proposed samples during the training process.

2.2 STRAIGHT-THROUGH GUMBEL-SOFTMAX ESTIMATOR

Continuous relaxations of one-hot vectors are suitable for problems such as learning hidden representations and sequence modeling. For scenarios in which we are constrained to sampling discrete values (e.g. from a discrete action space for reinforcement learning, or quantized compression), we discretize y using $\arg \max$ but use our continuous approximation in the backward pass by approximating $\nabla_{\theta} z \approx \nabla_{\theta} y$. We call this the Straight-Through (ST) Gumbel Estimator, as it is reminiscent of the biased path derivative estimator described in Bengio et al. (2013). ST Gumbel-Softmax allows samples to be sparse even when the temperature τ is high.

3 RELATED WORK

In this section we review existing stochastic gradient estimation techniques for discrete variables (illustrated in Figure 2). Consider a stochastic computation graph (Schulman et al., 2015) with discrete random variable z whose distribution depends on parameter θ , and cost function $f(z)$. The objective is to minimize the expected cost $L(\theta) = \mathbb{E}_{z \sim p_{\theta}(z)}[f(z)]$ via gradient descent, which requires us to estimate $\nabla_{\theta} \mathbb{E}_{z \sim p_{\theta}(z)}[f(z)]$.

3.1 PATH DERIVATIVE GRADIENT ESTIMATORS

For distributions that are reparameterizable, we can compute the sample z as a deterministic function g of the parameters θ and an independent random variable ϵ , so that $z = g(\theta, \epsilon)$. The path-wise gradients from f to θ can then be computed without encountering any stochastic nodes:

$$\frac{\partial}{\partial \theta} \mathbb{E}_{z \sim p_{\theta}}[f(z)] = \frac{\partial}{\partial \theta} \mathbb{E}_{\epsilon} [f(g(\theta, \epsilon))] = \mathbb{E}_{\epsilon \sim p_{\epsilon}} \left[\frac{\partial f}{\partial g} \frac{\partial g}{\partial \theta} \right] \quad (4)$$

For example, the normal distribution $z \sim \mathcal{N}(\mu, \sigma)$ can be re-written as $\mu + \sigma \cdot \mathcal{N}(0, 1)$, making it trivial to compute $\partial z / \partial \mu$ and $\partial z / \partial \sigma$. This reparameterization trick is commonly applied to training variational autoencoders with continuous latent variables using backpropagation (Kingma & Welling, 2013; Rezende et al., 2014b). As shown in Figure 2, we exploit such a trick in the construction of the Gumbel-Softmax estimator.

Biased path derivative estimators can be utilized even when z is not reparameterizable. In general, we can approximate $\nabla_{\theta} z \approx \nabla_{\theta} m(\theta)$, where m is a differentiable proxy for the stochastic sample. For Bernoulli variables with mean parameter θ , the Straight-Through (ST) estimator (Bengio et al., 2013) approximates $m = \mu_{\theta}(z)$, implying $\nabla_{\theta} m = 1$. For $k = 2$ (Bernoulli), ST Gumbel-Softmax is similar to the slope-annealed Straight-Through estimator proposed by Chung et al. (2016), but uses a softmax instead of a hard sigmoid to determine the slope. Rolfe (2016) considers an alternative approach where each binary latent variable parameterizes a continuous mixture model. Reparameterization gradients are obtained by backpropagating through the continuous variables and marginalizing out the binary variables.

One limitation of the ST estimator is that backpropagating with respect to the sample-independent mean may cause discrepancies between the forward and backward pass, leading to higher variance.